

Name: _____

Double Award Science – Chemistry Unit 1 Higher Tier

Once you have completed a paper and have marked it, fill out the table below, using the grade boundaries provided to add your grade. Then list which questions you did well and which questions you struggled with. For the questions you struggled with, review why and at the reasons. The most common reasons are

- Not understanding basic concepts (e.g. simple definitions)
- Calculation errors (forgetting to square numbers, etc.)
- Not reading questions carefully
- Not answering the full question (describe vs. explain)
- Laboratory/practical setup mistakes

For these questions it is also useful add notes to the question to help you next time and take a photo of these questions. Use these then to guide further revision.

Year	Score /60	Grade	Good Questions	Bad Questions	Reasons
2017					
2018					
2019					
2022					
2023					

Grade boundaries - Raw Score

Year	E	D	C	B	A	A*	Max
2017	6	9	15	21	28	35	42
2018	4	7	13	21	30	39	48
2019	4	8	13	19	25	31	37
2022	4	6	13	20	27	34	41
2023	5	7	11	18	26	34	42

Surname	Centre Number	Candidate Number
Other Names		0

GCSE – **NEW**

3430UB0-1



S17-3430UB0-1

SCIENCE (Double Award)**Unit 2: CHEMISTRY 1
HIGHER TIER**

FRIDAY, 16 JUNE 2017 – MORNING

1 hour 15 minutes

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	9	
2.	6	
3.	8	
4.	11	
5.	15	
6.	5	
7.	6	
Total	60	

3430UB01
01**ADDITIONAL MATERIALS**

In addition to this examination paper you will need a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen. Do not use correction fluid.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the continuation page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question **7** is a quality of extended response (QER) question where your writing skills will be assessed.

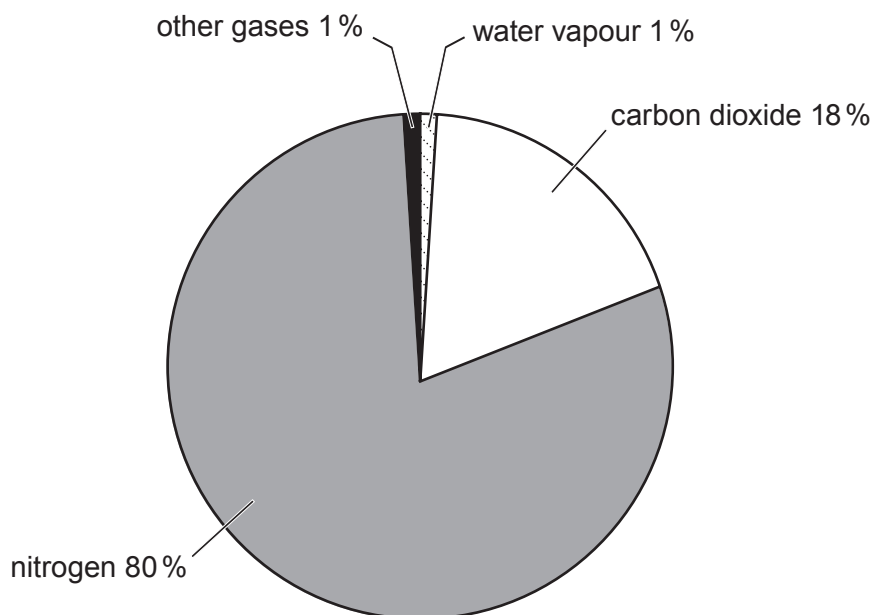
The Periodic Table is printed on the back cover of this paper and the formulae for some common ions on the inside of the back cover.



JUN173430UB0101

Answer all questions.

1. (a) The Earth's atmosphere has evolved over millions of years to its present composition. The pie chart shows the composition of the atmosphere at one stage of its evolution.



- (i) Describe the **main** differences in the composition of the atmosphere shown in the pie chart and that present on the Earth today. [3]

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- (ii) **Complete the table** to describe simple chemical tests that can be used in the laboratory to identify hydrogen and carbon dioxide. [2]

Gas	Test carried out	Expected observation
hydrogen		
carbon dioxide		



- (b) During the last 250 years the level of carbon dioxide in the atmosphere has increased. Most scientists believe that the increase in the level of carbon dioxide in the atmosphere has resulted in global warming.

The table below shows some data about the atmosphere between 1750 and 2000.

Year	Concentration of carbon dioxide in the atmosphere (parts per million)	Average global temperature (°C)
1750	278	13.3
1800	282	13.4
1850	288	13.5
1900	297	13.7
1950	310	14.0
2000	368	14.6

Use the data in the table to answer parts (i) and (ii).

- (i) Describe the trend in average global temperature between 1750 and 2000. [2]

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- (ii) Calculate the percentage increase in the concentration of carbon dioxide in the atmosphere from 1750 to 1850. [2]

percentage increase = %

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2. (a) The following table gives information about some atoms, **A-E**.

A-E are not the chemical symbols of the elements.

Atom	Number of protons	Number of neutrons
A	12	12
B	10	10
C	6	8
D	11	12
E	6	6

- (i) Give the letter, **A-E**, of the atom with a mass number of 12. [1]

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- (ii) Give the letter, **A-E**, of the atom found in Group 1 of the Periodic Table. [1]

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- (iii) Choose the letters, **A-E**, of the atoms that are isotopes. Give the reason for your choice. [2]

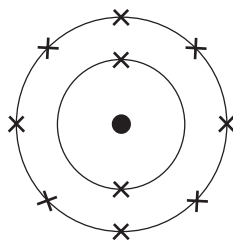
Letters and

Reason

.....



(b) The diagram below shows the electronic structure of an element in the Periodic Table.



(i) Using x to represent an electron, draw a diagram to show the electronic structure of the element which lies directly **below** this one in the Periodic Table. [1]

(ii) Explain why both of these elements are chemically unreactive. [1]

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3. The following information is about hardness in water.

Hardness in water is caused by the presence of dissolved calcium (Ca^{2+}) and magnesium (Mg^{2+}) ions. These ions come from compounds that are dissolved by the water as it passes through rocks. Because they are dissolved, these ions are not removed during the water treatment processes.

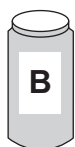
Hardness in water can be identified using soap solution. The relative hardness in different water samples can be determined by measuring the volume of soap solution required to produce a given lather.

Temporary hardness in water can be removed simply by boiling. Permanent hardness can be removed by passing the sample through an ion exchange column.

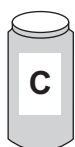
Three samples of water, **A**, **B** and **C**, were tested for hardness using soap solution. Distilled water was also tested.



Sample **A** required 8 cm^3 of soap solution to form lather before boiling.
It still required 8 cm^3 of soap solution to form lather after boiling.



Sample **B** required 12 cm^3 of soap solution to form lather before boiling.
It only required 6 cm^3 of soap solution to form lather after boiling.



Sample **C** required 15 cm^3 of soap solution to form lather before boiling.
It only required 2 cm^3 of soap solution to form lather after boiling.

Distilled water required 2 cm^3 of soap solution to form lather both before and after boiling.



- (a) Which **one** of these statements best describes the water samples before boiling? Tick (✓) the correct answer. [1]

samples **A**, **B** and **C** contain magnesium or calcium ions ☐

only samples **A** and **B** contain magnesium or calcium ions ☐

only samples **B** and **C** contain magnesium or calcium ions ☐

none of the samples contain magnesium or calcium ions ☐

- (b) What is the purpose of testing the distilled water? Tick (✓) the correct answer. [1]

to show the expected result for permanent hard water ☐

to show the expected result for temporary hard water ☐

to act as a control for the experiment ☐

to make sure the investigation is a fair test ☐

- (c) In carrying out the tests on the samples, which variables need to be kept the same each time? Tick (✓) the correct answer. [1]

volume of soap solution and number of times shaken ☐

volume of water sample and volume of soap solution used ☐

volume of water sample and number of times shaken ☐

volume of water sample, volume of soap solution and number of times shaken ☐

- (d) Which **one** of these statements best describes water sample **B**? Tick (✓) the correct answer and explain your choice. [3]

sample **B** contains temporary hardness only ☐

sample **B** contains permanent hardness only ☐

sample **B** contains a mixture of temporary and permanent hardness ☐

sample **B** does not contain hardness ☐

Explanation

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(e) In your opinion, do the advantages of living in a hard water area outweigh the disadvantages? Give **two** reasons to support your answer. [2]

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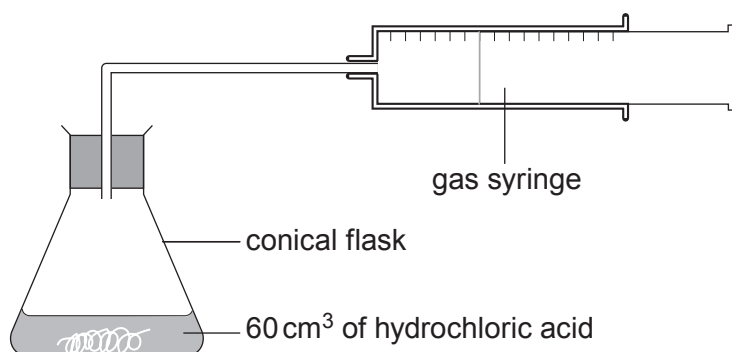
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4. Magnesium ribbon was reacted with excess dilute hydrochloric acid. The volume of hydrogen produced during the reaction was measured over time using a gas syringe. The experiment was carried out at a temperature of 20 °C.

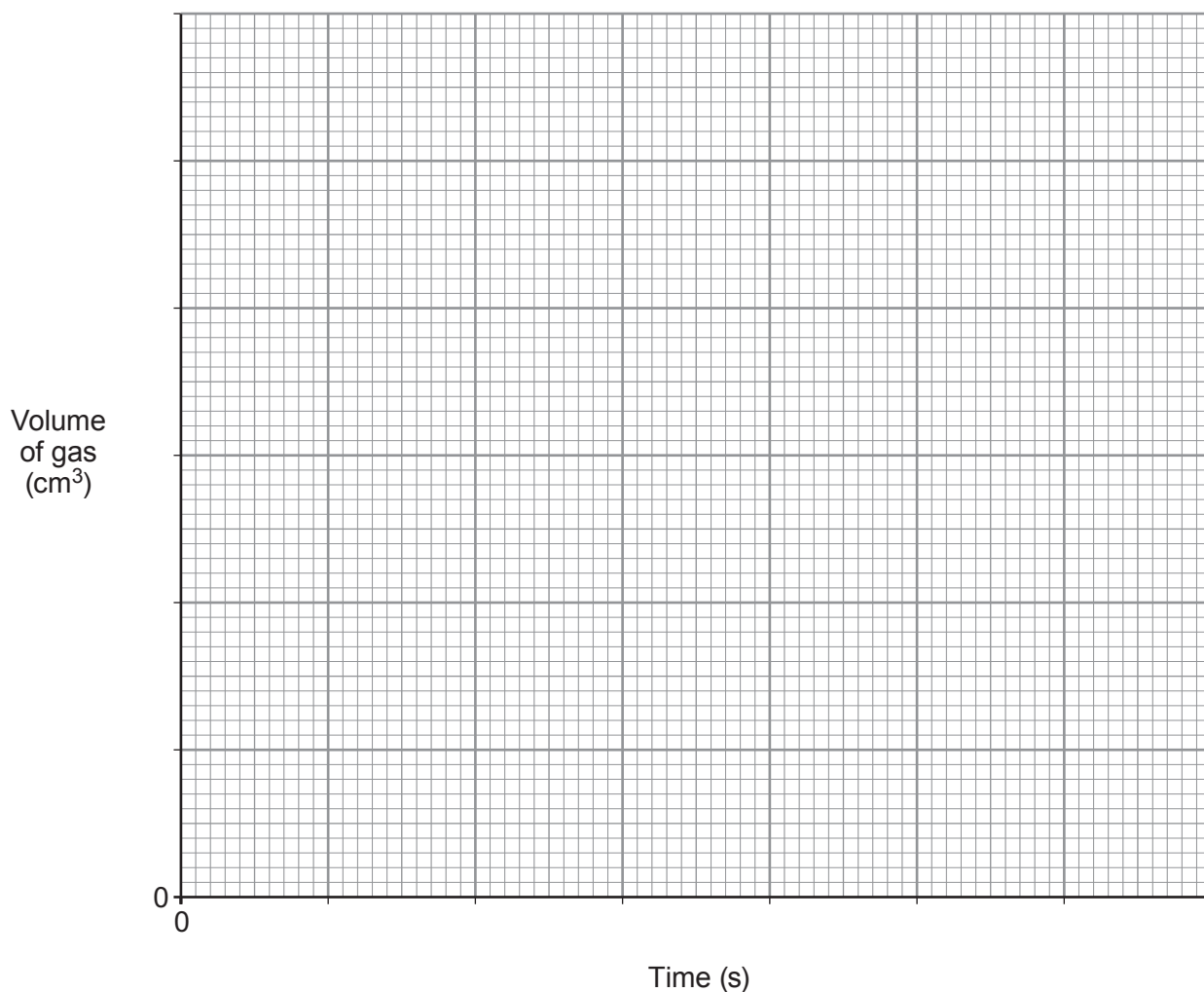


The table shows the results obtained in the experiment.

Time (s)	Volume of gas (cm ³)
0	0
10	34
20	64
30	78
40	94
50	98
60	100
70	100

- (a) Plot the data **on the grid opposite** and draw a suitable line. Label the graph **A**. [3]
- (b) **On the same grid**, sketch the graph that would be obtained if a piece of magnesium ribbon of half the length were reacted with another 60 cm³ of the same hydrochloric acid. Label this graph **B**. [2]





- (c) If the original experiment were repeated at a higher temperature, the reaction would occur at a higher rate. Explain this using particle theory. [2]

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- (d) This method can be used to investigate the rate of this reaction. Without changing the method or apparatus used, state what should be done to improve the overall strength of the evidence collected. [2]

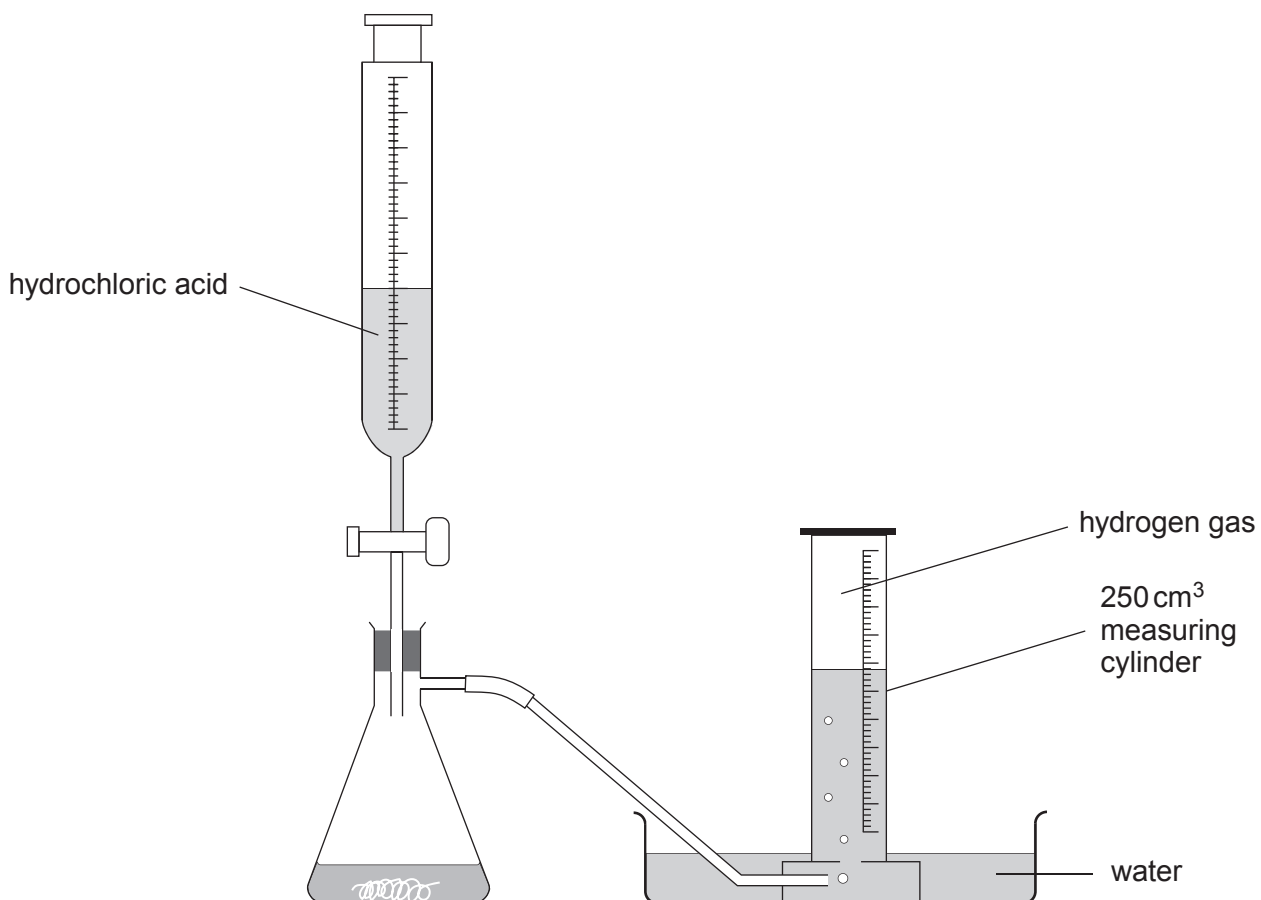
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- (e) Give an advantage and a disadvantage of using the following alternative apparatus to collect the hydrogen gas. [2]



Advantage

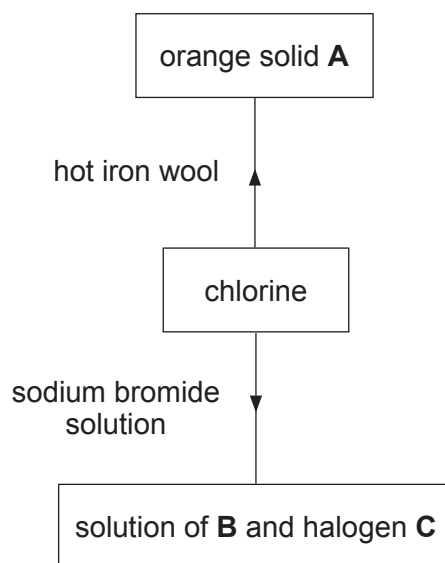
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Disadvantage

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5. (a) The diagram shows some reactions of chlorine.



- (i) Give the chemical **formulae** for substances **A**, **B** and **C**. [3]

A

B

C

- (ii) State, giving a reason, the **main** safety precaution that must be taken when carrying out these reactions. [1]

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- (iii) State and explain how the observations for the reaction with sodium bromide would be different if the chlorine were replaced with iodine. [3]

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- (b) Chlorine can form a number of compounds by reacting with oxygen gas. One of these is used in water treatment.

A sample of this compound contains 0.71 g of chlorine and 0.64 g of oxygen. Calculate its simplest formula. [3]

Simplest formula

- (c) Silver nitrate solution is used to detect the presence of halide ions in solution.

- (i) The equation below represents the reaction between silver nitrate solution and potassium bromide solution.



Write the **ionic** equation for the reaction. **Include state symbols.** [2]

-
- (ii) Write the balanced **symbol** equation for the reaction between calcium iodide solution and silver nitrate solution. [3]

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6. The balanced symbol equation for the reaction between sodium and oxygen is given below.



- (a) Calculate the mass of sodium oxide that should be produced when 0.736 g of sodium is burned in air. [3]

mass = g

- (b) When the reaction was carried out using 0.736 g of sodium, it was found that the mass of sodium oxide formed was 0.338 g.

- (i) Use your answer to part (a) to calculate the percentage yield of the reaction. [1]

percentage yield = %

- (ii) If no product was lost, suggest **one** possible explanation for the percentage yield not being 100%. [1]

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7. Describe how the reactions of lithium, sodium and potassium with water can be used to show the trend in reactivity in Group 1. Explain this trend in terms of the electronic structures of the elements. [6 QER]

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END OF PAPER



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^-
ammonium	NH_4^+	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^-
calcium	Ca^{2+}	fluoride	F^-
copper(II)	Cu^{2+}	hydroxide	OH^-
hydrogen	H^+	iodide	I^-
iron(II)	Fe^{2+}	nitrate	NO_3^-
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^+	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^+		
silver	Ag^+		
sodium	Na^+		
zinc	Zn^{2+}		



2
1



4

9

4

O

7 Li Lithium 3	9 Be Beryllium 4
23 Na Sodium 11	24 Mg Magnesium 12
39 K Potassium 19	40 Ca Calcium 20
86 Rb Rubidium 37	88 Sr Strontium 38
133 Cs Caesium 55	137 Ba Barium 56
223 Fr Francium 87	226 Ra Radium 88

						4 He Helium
11 B Boron	12 C Carbon	14 N Nitrogen	16 O Oxygen	19 F Fluorine	20 Ne Neon	
27 Al Aluminium	28 Si Silicon	31 P Phosphorus	32 S Sulfur	35.5 Cl Chlorine	40 Ar Argon	
70 Ga Gallium	73 Ge Germanium	75 As Arsenic	79 Se Selenium	80 Br Bromine	84 Kr Krypton	
115 In Indium	119 Sn Tin	122 Sb Antimony	128 Te Tellurium	127 I Iodine	131 Xe Xenon	
204 Tl Thallium	207 Pb Plead	209 Bi Bismuth	210 Po Polonium	210 At Astatine	222 Rn Radon	

48	51	52	55	56	59	59	63.5	65
Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
Titanium	Vanadium	Chromium	Manganese	Iron	Cobalt	Nickel	Copper	Zinc
22	23	24	25	26	27	28	29	30
91	93	96	99	101	103	106	108	112
Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd
Zirconium	Niobium	Molybdenum	Technetium	Ruthenium	Rhodium	Palladium	Silver	Cadmium
40	41	42	43	44	45	46	47	48
179	181	184	186	190	192	195	197	201
Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg
Hafnium	Tantalum	Tungsten	Rhenium	Osmium	Iridium	Platinum	Gold	Mercury
72	73	74	75	76	77	78	79	80

223 Fr Francium 87	226 Ra Radium 88	227 Ac Actinium 89
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Key

relative atomic mass

Symbol	Name	Z
A_r		

atomic number

Surname	Centre Number	Candidate Number
Other Names		0

GCSE – **NEW**

3430UB0-1



S18-3430UB0-1

SCIENCE (Double Award)**Unit 2: CHEMISTRY 1
HIGHER TIER****WEDNESDAY, 13 JUNE 2018 – MORNING****1 hour 15 minutes**

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	8	
2.	7	
3.	6	
4.	8	
5.	5	
6.	6	
7.	11	
8.	9	
Total	60	

3430UB01
01**ADDITIONAL MATERIALS**

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The Periodic Table is printed on the back cover of this paper and the formulae for some common ions on the inside of the back cover.



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Answer all questions.

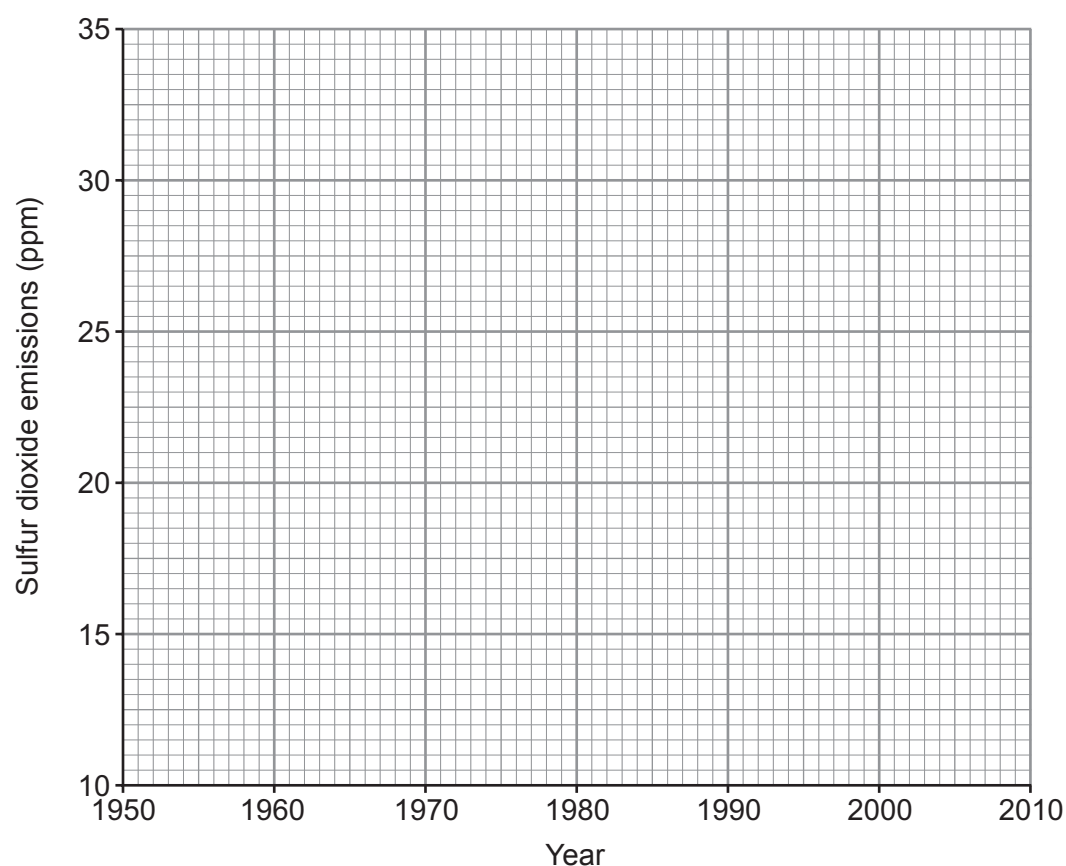
1. Burning fossil fuels containing sulfur causes sulfur dioxide, SO_2 , to be released into the atmosphere.

The table shows sulfur dioxide emissions in the UK between 1950 and 2010.

Year	Sulfur dioxide emissions (ppm)
1950	12.0
1960	16.0
1970	21.5
1980	29.5
1990	29.0
2000	24.0
2010	18.5

ppm = parts per million

- (a) (i) On the grid plot the sulfur dioxide emissions against the year and draw a suitable line. [3]



- (ii) Describe how sulfur dioxide emissions changed between 1950 and 2010. [2]

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- (iii) The UK government introduced a regulation to reduce sulfur dioxide emissions in the 1980s. From your graph, state why it is difficult to decide exactly the year when the regulation came into force. [1]

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- (b) Sulfur dioxide can be converted to sulfur and water by reacting it with hydrogen sulfide, H_2S .

Complete and balance the symbol equation for this reaction. [2]



2. The table shows some physical properties of Group 6 elements.

Element	Melting point (°C)	Boiling point (°C)	Density (g/cm ³)	Electrical conductor
oxygen	−219	−183	0.0014	no
sulfur	115	445	2.0	no
selenium	221	685	4.8	semi-conductor
tellurium	450	988	6.2	semi-conductor

(a) (i) Describe the trend in the melting points of the Group 6 elements. [1]

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(ii) Give the physical state of selenium at 400 °C. Give a reason for your choice. [2]

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(iii) Explain why it is difficult to classify selenium as either a metal or a non-metal. [1]

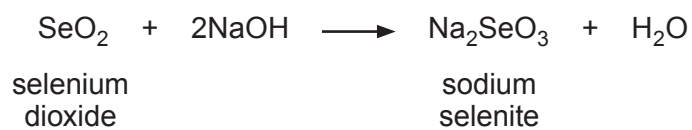
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- (b) Selenium dioxide reacts with sodium hydroxide to produce sodium selenite and water.



- (i) Calculate the relative formula mass (M_r) of sodium selenite, Na_2SeO_3 . [1]

$$A_r(\text{Na}) = 23 \qquad A_r(\text{Se}) = 79 \qquad A_r(\text{O}) = 16$$

$$M_r = \dots\dots\dots$$

- (ii) Calculate the percentage by mass of selenium in sodium selenite. [2]

$$\text{Percentage} = \dots\dots\dots \%$$



3. (a) The table gives the composition of six particles, **A-F**, which are either atoms or ions.

Particle	Number of protons	Number of neutrons	Number of electrons
A	14	14	14
B	19	20	18
C	15	16	18
D	16	16	16
E	11	12	11
F	12	12	10

- (i) Which particles are atoms? Explain your choice.

[2]

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- (ii) Which particles are positive ions? Give the charges on the particles you have chosen.

[2]

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- (b) Carbon has two isotopes – carbon-12 and carbon-14.

Using these examples, explain what is meant by the term *isotope*.

[2]

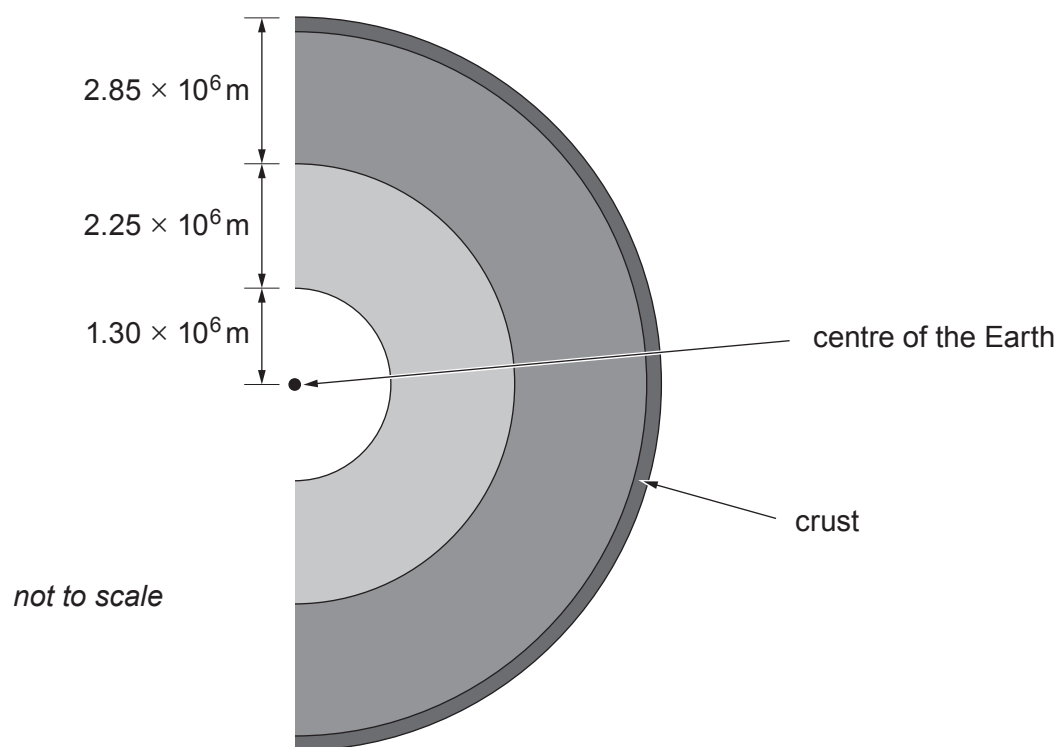
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4. The Earth is almost spherical. The diagram shows a section of the layered structure of the Earth.



- (a) The thickness of the crust at one location is 0.97 % of the distance from the surface to the centre of the Earth.

Use this information to calculate the thickness of the crust. Give your answer in **standard form**. [3]

Thickness of crust = m



(b) (i) Describe the theory of plate tectonics.

[3]

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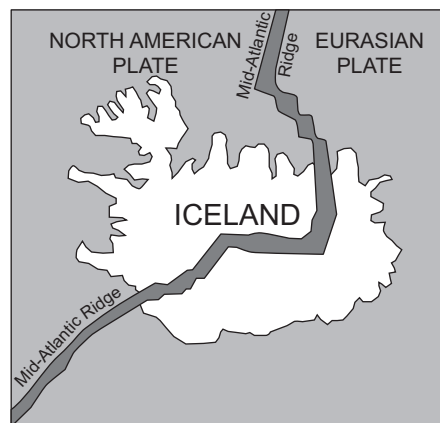
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(ii) Iceland lies on the Mid-Atlantic Ridge which is formed at a constructive plate boundary. Explain the formation of the Mid-Atlantic Ridge.

[2]



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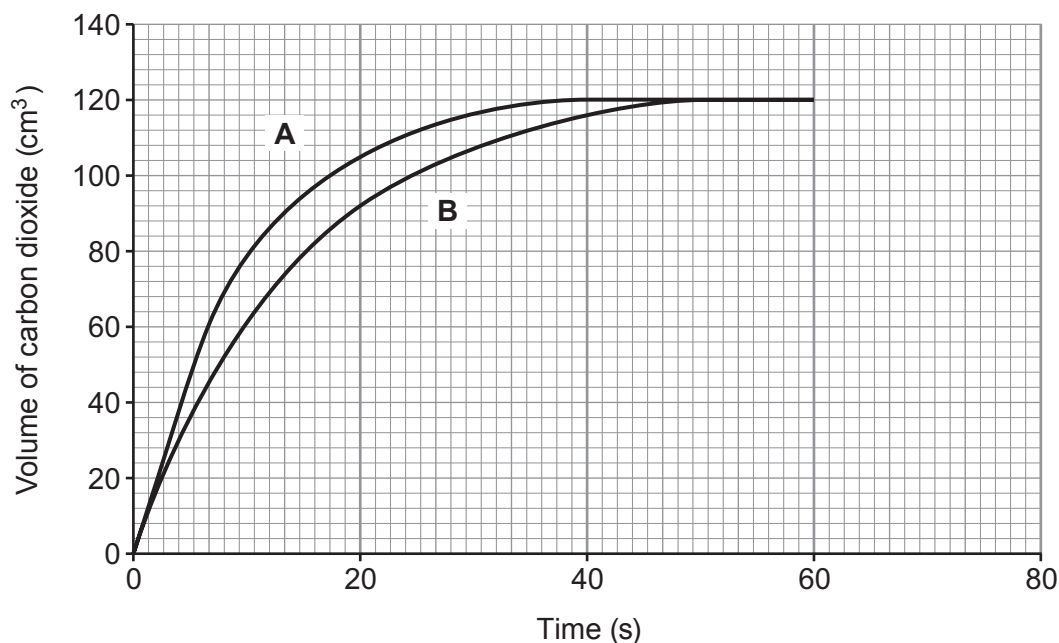
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5. The rate of reaction between hydrochloric acid and calcium carbonate was studied. Two experiments, **A** and **B**, were carried out. The same concentration of acid and the same mass of calcium carbonate were used, with the acid in excess each time. The volume of gas produced was measured for 60 seconds. The results of these experiments are shown in the graphs.



- (a) Using the particle theory, state and explain **two** factors that may be responsible for the higher rate seen in experiment **A**. [4]

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- (b) State how the graphs show that the same mass of calcium carbonate was used in both experiments. [1]

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6. Explain the trends in reactivity of the elements in Groups 1 and 7 of the Periodic Table. [6 QER]

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7. (a) (i) A group of students were given three water samples. One was soft water, one was temporary hard water and one was permanent hard water.

Describe a method that the students could use to find out which is which. Give the expected results. You do **not** need to include the detail required to ensure a fair test. [4]

- (ii) The samples from part (i) were labelled **X**, **Y** and **Z** and sent to have their ion content measured. The table shows the results.

Ions present	Ion content in water sample (mg/dm ³)		
	X	Y	Z
sodium	29	116	23
potassium	12	15	11
magnesium	31	4	98
calcium	141	2	27
hydrogencarbonate	30	19	219
chloride	17	14	20
sulfate	346	6	27
nitrate	12	15	19

Use this information to find whether sample **Z** is the soft water, the temporary hard water or the permanent hard water. Give a reason for your choice. [2]



- (b) (i) Permanent hard water can be softened by ion exchange. Explain how ion exchange works. [2]

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- (ii) Explain why the resin in ion exchange stops working after continued use. [1]

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- (c) A litre bottle of mineral water contains 184 mg of dissolved calcium sulfate, CaSO_4 . Calculate the number of moles of calcium sulfate present.

Give your answer to **three** significant figures.

$$1 \text{ mg} = 0.001 \text{ g} \quad [2]$$

$$A_r(\text{Ca}) = 40 \quad A_r(\text{S}) = 32 \quad A_r(\text{O}) = 16$$

Moles of calcium sulfate = mol



8. The most common understanding of the term 'heavy metal' is a metallic element which is toxic and has a high density, atomic number and relative atomic mass. The definitions used vary depending on the context. In metallurgy, for example, a heavy metal is defined on the basis of density, whereas in physics the distinguishing factor is atomic number. A chemist would likely be more concerned with chemical behaviour. More specific definitions have been published but none of these have been widely accepted.

Despite this lack of agreement, the term is widely used in science. Heavy metals are sometimes defined as metals with a density greater than 5 g/cm^3 . They are often highly toxic or damaging to the environment. Chromium, arsenic, cadmium, mercury and lead have the greatest potential to cause harm on account of their extensive use.

Heavy metals are dangerous because they tend to bio-accumulate. Bio-accumulation occurs when the toxic chemical is taken into the body faster than it can be excreted. Lead can have an adverse impact on mental development in infants and children. Lead may also be a factor in behavioural problems. Heavy metal poisoning could result, for instance, from drinking-water contamination.

Lead is the most prevalent heavy metal contaminant. As a component of tetraethyl lead, $(\text{CH}_3\text{CH}_2)_4\text{Pb}$, it was used extensively in 'leaded petrol' from the 1930s-1970s. Although the use of leaded petrol has been phased out, soils next to roads can have high lead concentrations – see **Figure 1**. Lead-based paints were another early source of lead pollution but their use is now banned in the UK. **Figure 2** opposite shows how the amount of lead used in paint and petrol in the USA changed over the 20th century.

Figure 1

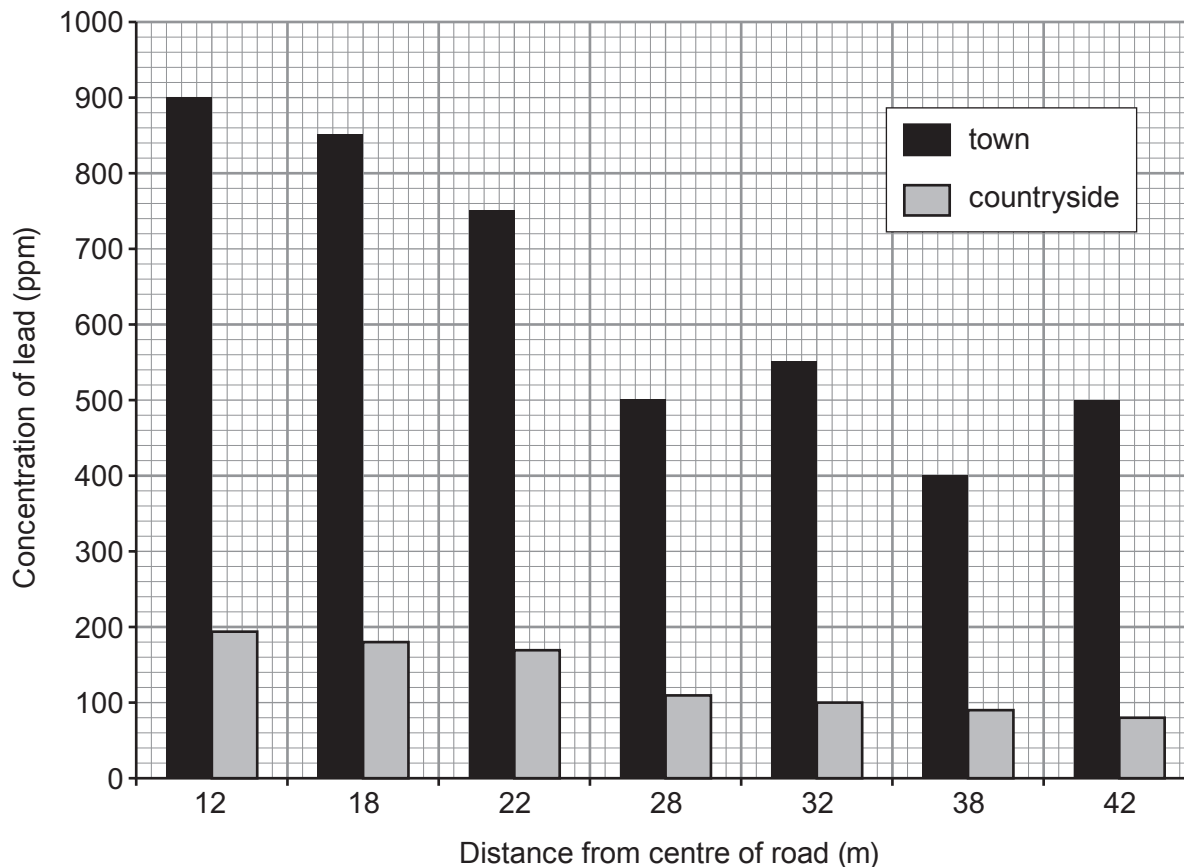
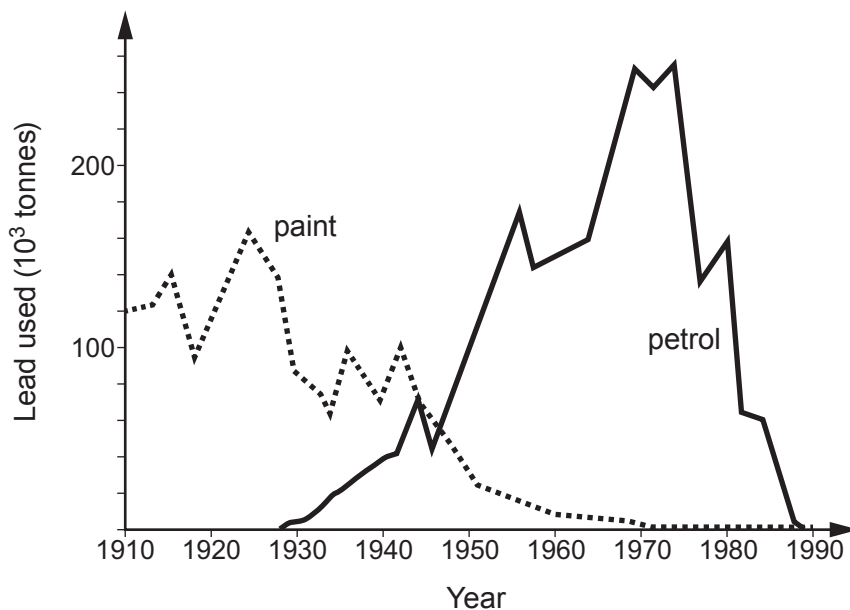


Figure 2



- (a) (i) Put a tick (✓) in the box next to the statement which **best** describes why a single definition of a 'heavy metal' is not accepted by all scientists. [1]

metallurgists, physicists and chemists do not trust each other

☐

metallurgists, physicists and chemists are not concerned with the fact that heavy metals are toxic

☐

metallurgists, physicists and chemists are concerned with different properties of heavy metals

☐

metallurgists, physicists and chemists study different heavy metals

☐

- (ii) Since the 1970s the use of lead water pipes has been prohibited across Europe. Explain why this is the case. [1]

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- (iii) Put a tick (✓) in the box next to the **two** statements which describe the conclusions that can be drawn from the data in **Figure 1**. [2]

lead contamination in road-side soil decreases in towns and in the countryside as the distance from the centre of the road increases ☐

the decrease in lead contamination between 12 m and 42 m from the centre of the road is greater in towns than in the countryside ☐

lead contamination in road-side soil at all distances is much greater in towns than in the countryside ☐

lead contamination in road-side soil decreases between 12 m and 42 m from the centre of the road in the countryside ☐

there is approximately 50 % more lead contamination in road-side soil in towns compared to the countryside between 12 m and 28 m from the centre of the road ☐

there is an overall decrease of nearly 70 % in the lead contamination of road-side soil in towns between 12 m and 42 m from the centre of the road ☐

- (iv) In the mid-1970s the use of lead in paints and the manufacture of cars that used leaded petrol was banned. Put a tick (✓) in the box next to the statement which **best** describes what happened after the ban. [1]

paint and petrol were lead-free by the mid-1970s anyway ☐

paint and petrol were not lead-free until 1980 ☐

the use of lead-based paint and leaded petrol increased until 1980 before decreasing ☐

it took 15 years for paint and petrol to become lead-free ☐



- (b) (i) Lead can form a number of oxides with different formulae. Red lead, Pb_3O_4 , is used to make batteries. It is manufactured by reacting lead carbonate with oxygen. Carbon dioxide is also produced.

Balance the equation for this reaction.

[1]



- (ii) 11.36 g of a lead oxide contains 10.18 g of lead. Calculate the empirical formula of this oxide.

[3]

$$A_r(\text{Pb}) = 207 \quad A_r(\text{O}) = 16$$

Empirical formula

9

END OF PAPER



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^-
ammonium	NH_4^+	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^-
calcium	Ca^{2+}	fluoride	F^-
copper(II)	Cu^{2+}	hydroxide	OH^-
hydrogen	H^+	iodide	I^-
iron(II)	Fe^{2+}	nitrate	NO_3^-
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^+	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^+		
silver	Ag^+		
sodium	Na^+		
zinc	Zn^{2+}		





THE PERIODIC TABLE

Group

1 2

3 4 5 6 7 0

1	H	Hydrogen	1
---	---	----------	---

7	Li	Lithium	3
9	Be	Beryllium	4
23	Na	Sodium	11
24	Mg	Magnesium	12
39	K	Potassium	19
40	Ca	Calcium	20
86	Rb	Rubidium	37
88	Sr	Strontium	38
133	Cs	Caesium	55
223	Fr	Francium	87
45	Sc	Scandium	21
48	Ti	Titanium	22
51	V	Vanadium	23
52	Cr	Chromium	24
55	Mn	Manganese	25
56	Fe	Iron	26
59	Co	Cobalt	27
59	Ni	Nickel	28
63.5	Cu	Copper	29
65	Zn	Zinc	30
89	Y	Yttrium	39
89	Sc	Scandium	21
91	Zr	Zirconium	40
93	Nb	Niobium	41
96	Mo	Molybdenum	42
99	Tc	Technetium	43
101	Ru	Ruthenium	44
103	Rh	Rhodium	45
106	Pd	Palladium	46
108	Ag	Silver	47
112	Cd	Cadmium	48
115	In	Indium	49
73	Ge	Germanium	32
75	As	Arsenic	33
79	Se	Selenium	34
80	Br	Bromine	35
84	Kr	Krypton	36
137	Ba	Barium	56
139	La	Lanthanum	57
226	Ra	Radium	88
227	Ac	Actinium	89

11	B	Boron	5
12	C	Carbon	6
14	N	Nitrogen	7
16	O	Oxygen	8
19	F	Fluorine	9
20	Ne	Neon	10
27	Al	Aluminium	13
28	Si	Silicon	14
31	P	Phosphorus	15
32	S	Sulfur	16
35.5	Cl	Chlorine	17
40	Ar	Argon	18
70	Ga	Gallium	31
73	Ge	Germanium	32
75	As	Arsenic	33
79	Se	Selenium	34
80	Br	Bromine	35
84	Kr	Krypton	36
115	In	Indium	49
119	Sn	Tin	50
122	Sb	Antimony	51
127	I	Iodine	53
131	Xe	Xenon	54
204	Tl	Thallium	81
207	Pb	Lead	82
209	Bi	Bismuth	83
210	Po	Polonium	84
210	At	Astatine	85
222	Rn	Radon	86

Key

Ar	relative atomic mass
Symbol	
Name	
Z	atomic number

Surname	Centre Number	Candidate Number
Other Names		0

**GCSE****3430UB0-1**

S19-3430UB0-1

WEDNESDAY, 12 JUNE 2019 – MORNING**SCIENCE (Double Award)****Unit 2: CHEMISTRY 1
HIGHER TIER****1 hour 15 minutes****ADDITIONAL MATERIALS**

In addition to this examination paper you will need a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen. Do not use correction fluid.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question **6** is a quality of extended response (QER) question where your writing skills will be assessed.

The Periodic Table is printed on the back cover of this paper and the formulae for some common ions on the inside of the back cover.

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	8	
2.	7	
3.	8	
4.	9	
5.	5	
6.	6	
7.	9	
8.	8	
Total	60	

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Answer all questions.

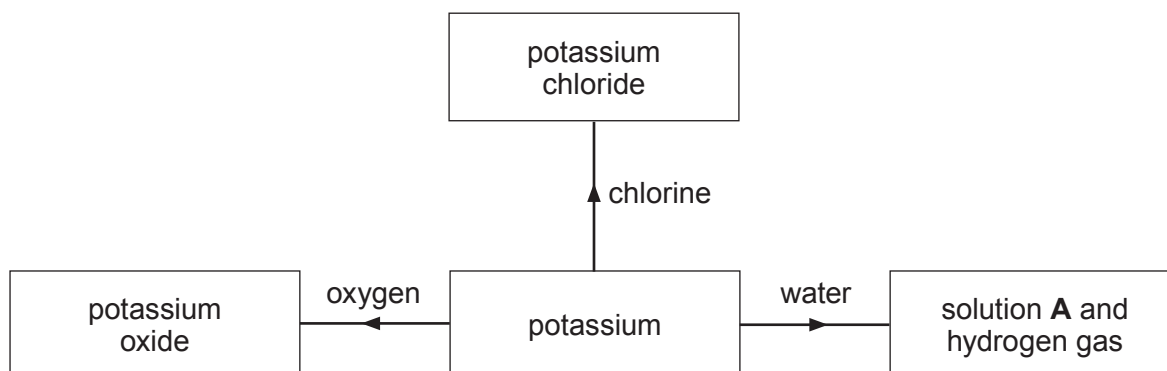
1. (a) The table gives information about some elements.

Element	Electronic structure	Group	Period
oxygen	2,6	6	2
chlorine		7	3
.....	2,8,5	5	3
potassium	2,8,8,1	1	

Complete the table.

[3]

- (b) The flow chart shows some of the reactions of potassium.



- (i) State **one** observation you would make when potassium reacts with water. [1]

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- (ii) Apart from wearing gloves and safety goggles, give **one** safety precaution that should be taken when adding potassium to water. [1]

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(iii) Give the formula of solution **A**. [1]

.....

(iv) Suggest a value for the pH of solution **A**. [1]

.....

(v) Name a Group 1 metal that is **more** reactive than potassium. [1]

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8

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03



2. (a) The Earth's early atmosphere around 4 000 million years ago contained mainly carbon dioxide and water vapour produced by volcanoes.

- (i) Explain why the large percentage of water vapour in the Earth's atmosphere decreased over geological time. [2]

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- (ii) Give **two** reasons why the percentage of carbon dioxide in the Earth's atmosphere has decreased over geological time. [2]

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- (b) During the last 250 years the percentage of carbon dioxide in the Earth's atmosphere has increased from 0.03 % to 0.04 %. This has led to increased global warming. Give **one** reason for this increase and explain why global warming is a cause for concern. [2]

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- (c) Ammonia present in the Earth's early atmosphere reacted with oxygen to produce nitrogen and water vapour. Complete the balancing of the symbol equation for this reaction. [1]



3. Some Year 10 students were given three unknown white solids – **A**, **B** and **C**.

They carried out a series of flame tests and silver nitrate tests to identify the solids.

Their results are shown in the table.

Solid	Observations	
	Flame test	Silver nitrate test
A	apple-green flame	cream precipitate
B	red flame	white precipitate
C	yellow flame	yellow precipitate

- (a) Name solids **A**, **B** and **C**.

[3]

A

B

C

- (b) Complete and balance the symbol equation for the reaction between magnesium chloride and silver nitrate.

[2]



- (c) 0.103 g of silver nitrate, AgNO_3 , was used to make up a solution.

Calculate the number of moles of silver nitrate in this mass. Give your answer in **standard form**.

[3]

$$A_r(\text{Ag}) = 108$$

$$A_r(\text{N}) = 14$$

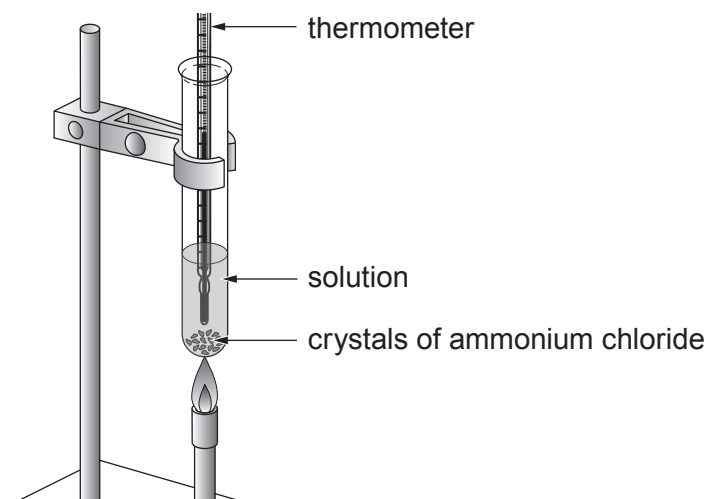
$$A_r(\text{O}) = 16$$

Number of moles = mol



4. A student investigates the solubility of ammonium chloride by adding different masses to 10 g of water.

He uses the apparatus shown.



10 g of water is placed in a boiling tube and 3.0 g of ammonium chloride is added.

The tube is heated until all the solid dissolves.

The tube is allowed to cool.

The temperature at which solid ammonium chloride first appears is recorded.

The experiment is repeated using different masses of ammonium chloride.

The results are shown in the table.

Mass of ammonium chloride in 10 g of water (g)	3.0	3.3	4.1	5.2	5.9	6.6
Temperature at which solid ammonium chloride first appears (°C)	4	10	30	52	68	80

- (a) What practical problem is the student likely to come across in finding the first two results? Suggest how this problem might be overcome. [2]

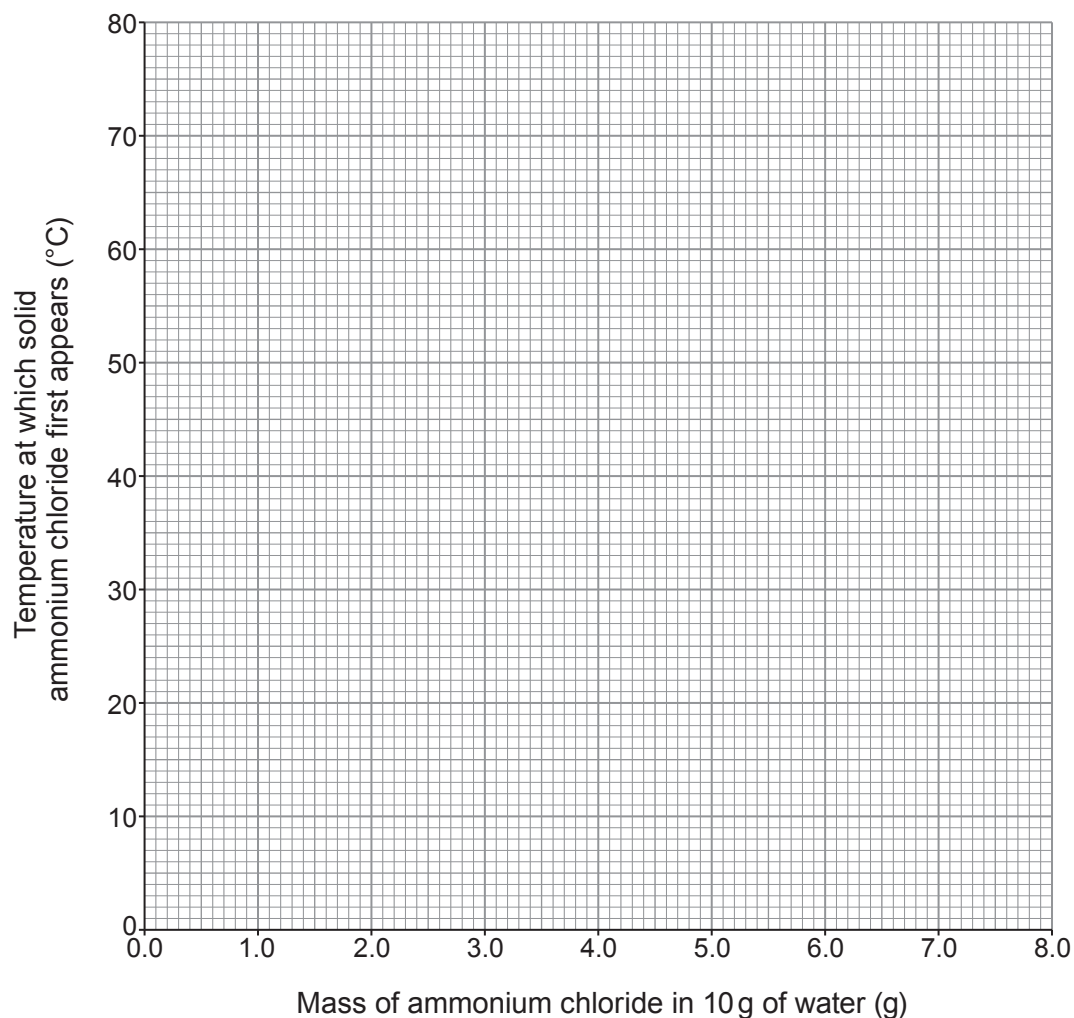
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- (b) (i) On the grid below, plot the temperature at which solid ammonium chloride first appears against the mass of ammonium chloride in 10 g of water. Draw a suitable line. [3]



- (ii) The student is given a boiling tube containing 5.0 g of ammonium chloride in 10 g of water. He stirs the ammonium chloride in the water and heats it to a temperature of 45°C.

State whether all the ammonium chloride dissolves. Give a reason for your answer. [1]

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- (c) The student is asked to use a **different** method to find the exact solubility of another compound in water at room temperature. He knows that it has a value of approximately 7g per 100g of water at this temperature.

He is given a 5.0g sample of the compound and common laboratory equipment but **no heating apparatus**.

Describe how he would carry out his method and how he would find the solubility. [3]

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5. (a) A bottle contains a mixture of liquids **E** and **F**. Liquid **E** has a boiling point of 57 °C and liquid **F** has a boiling point of 95 °C.

Describe the process of distillation and explain why it can be used to separate these liquids. [3]

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- (b) One molecule of liquid **E** contains **two** oxygen atoms. The percentage by mass of oxygen in liquid **E** is 43.2 %.

Use the following equation to calculate the relative molecular mass (M_r) of liquid **E**. [2]

$$\frac{\text{mass of oxygen}}{M_r} \times 100 = 43.2$$

$$A_r(\text{O}) = 16$$

$$M_r = \dots\dots\dots$$

5



Examiner
only

6. Explain how the processes of boiling and adding washing soda remove hardness from water. [6 QER]
Include equations in your answer.

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7. (a) (i) Chlorine is a non-metal found in Group 7 of the Periodic Table. When it is bubbled into a solution of potassium iodide there is a colour change from pale green to brown. Explain why this reaction occurs. [2]

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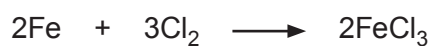
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- (ii) Write the balanced symbol equation for the reaction between chlorine and potassium iodide. [2]

.....

- (b) The symbol equation for the reaction between iron and chlorine is as follows.



Calculate the mass of chlorine needed to react with 1.32 g of iron. [3]

$$A_r(\text{Fe}) = 56$$

$$A_r(\text{Cl}) = 35.5$$

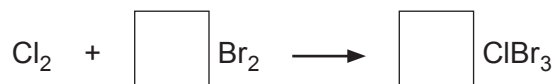
Mass of chlorine = g



- (c) (i) Under certain conditions, Group 7 elements will react with each other to produce new compounds.

When chlorine is reacted with bromine, chlorine tribromide is made.

Balance the symbol equation for this reaction. [1]



- (ii) A chemist calculated that if she reacted 7.00 g of chlorine with an excess of bromine, the theoretical mass of chlorine tribromide produced is 27.55 g.

However, when she carried out the experiment using 7.00 g of chlorine the mass of chlorine tribromide obtained was 21.34 g.

Calculate the percentage yield of chlorine tribromide. [1]

Percentage yield = %



8. Hydrogen peroxide solution, H_2O_2 , is used in commercial stain removers.

A GCSE class investigated how effective four stain removers are at removing stains. Stain removers **A**, **B**, **C** and **D** contain different concentrations of hydrogen peroxide.

The students tested how effective each one is at removing an identical oil stain from four towels.

Their findings are outlined below.

Stain remover A

Working temperature	= 50 °C
Cost per 100 cm ³	= 99p
Time to remove stain	= 40 min
Volume needed	= 20 cm ³

Stain remover B

Working temperature	= 30 °C
Cost per 100 cm ³	= £1.99
Time to remove stain	= 40 min
Volume needed	= 10 cm ³

Stain remover C

Working temperature	= 20 °C
Cost per 100 cm ³	= £2.49
Time to remove stain	= 20 min
Volume needed	= 5 cm ³

Stain remover D

Working temperature	= 30 °C
Cost per 100 cm ³	= £1.49
Time to remove stain	= 30 min
Volume needed	= 10 cm ³

- (a) In carrying out this investigation, which variables were kept the same in order to get valid results? Tick (✓) the correct answer. [1]

type of oil used, towel material and volume of hydrogen peroxide

☐

type of oil used, towel material and temperature of stain remover

☐

type of oil used and towel material

☐

type of oil used, towel material and cost of stain remover

☐


- (b) Tick (✓) **all** of the statements which **could** explain why stain remover **A** has to be heated to 50 °C before it removes the stain. [1]

it is the cheapest stain remover

☐

it is heat resistant

☐

it has a low concentration of hydrogen peroxide

☐

it takes a long time to work

☐

- (c) The students found that stain removers **B** and **D** used the same volume and worked best at the same temperature.

Assuming that they have the same hydrogen peroxide concentration, suggest a possible reason why **D** removes the stain more quickly than **B**. [1]

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(d) One student went on to investigate the decomposition of hydrogen peroxide.

The equation for the reaction is as follows.

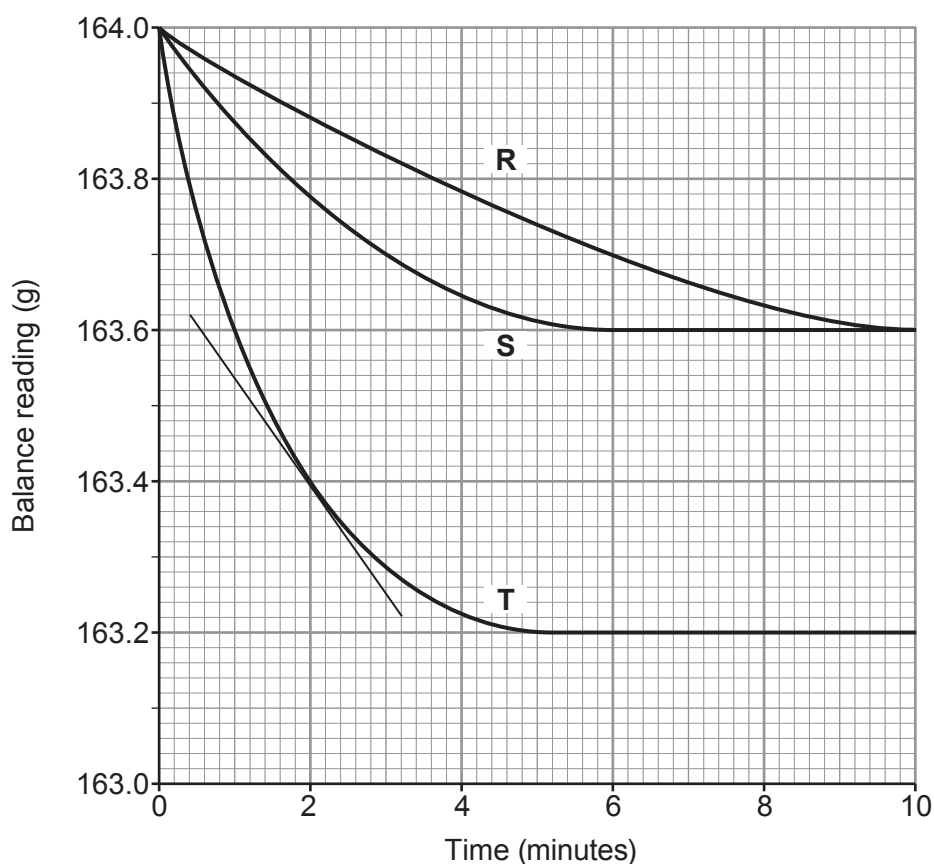


The student investigated the effect of changing the concentration of the hydrogen peroxide solution on the rate of the reaction. She used manganese dioxide as a catalyst in each experiment.

This is the method she used.

- Pour 50 cm³ of hydrogen peroxide solution of concentration **R** into a conical flask on a digital balance.
- Add 1 g of catalyst and place some cotton wool loosely in the neck of the flask. Record the balance reading and immediately start a stopwatch.
- Record the balance reading every minute until the mass no longer changes.
- Carry out the experiment twice more using hydrogen peroxide of different concentrations, **S** and **T**.

Her results are plotted on the grid below.



- (i) Using the tangent shown on the graph, calculate the rate of reaction for concentration **T** at 2 minutes. Show your working. [2]

Rate at 2 minutes = g/minute

- (ii) The initial rate for concentration **S** is half the initial rate for concentration **T**. Explain this difference in rate in terms of the particle theory. [3]

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8

END OF PAPER



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^-
ammonium	NH_4^+	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^-
calcium	Ca^{2+}	fluoride	F^-
copper(II)	Cu^{2+}	hydroxide	OH^-
hydrogen	H^+	iodide	I^-
iron(II)	Fe^{2+}	nitrate	NO_3^-
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^+	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^+		
silver	Ag^+		
sodium	Na^+		
zinc	Zn^{2+}		





THE PERIODIC TABLE

Group

1 2

3 4 5 6 7 0

1	H	Hydrogen	1
---	---	----------	---

7	Li	Lithium	3
9	Be	Beryllium	4
23	Na	Sodium	11
24	Mg	Magnesium	12
39	K	Potassium	19
40	Ca	Calcium	20
86	Rb	Rubidium	37
88	Sr	Strontium	38
133	Cs	Caesium	55
223	Fr	Francium	87

11	B	Boron	5
12	C	Carbon	6
14	N	Nitrogen	7
16	O	Oxygen	8
19	F	Fluorine	9
20	Ne	Neon	10
27	Al	Aluminium	13
28	Si	Silicon	14
31	P	Phosphorus	15
32	S	Sulfur	16
35.5	Cl	Chlorine	17
40	Ar	Argon	18

59	Co	Cobalt	27
56	Fe	Iron	26
55	Mn	Manganese	25
52	Cr	Chromium	24
51	V	Vanadium	23
48	Ti	Titanium	22
45	Sc	Scandium	21
89	Y	Yttrium	39
88	Sr	Strontium	38
91	Zr	Zirconium	40
93	Nb	Niobium	41
96	Mo	Molybdenum	42
99	Tc	Technetium	43
101	Ru	Ruthenium	44
103	Rh	Rhodium	45
106	Pd	Palladium	46
108	Ag	Silver	47
112	Cd	Cadmium	48
65	Zn	Zinc	30
63.5	Cu	Copper	29
70	Ga	Gallium	31
73	Ge	Germanium	32
75	As	Arsenic	33
79	Se	Selenium	34
80	Br	Bromine	35
84	Kr	Krypton	36
115	In	Indium	49
119	Sn	Tin	50
122	Sb	Antimony	51
128	Te	Tellurium	52
127	I	Iodine	53
131	Xe	Xenon	54
204	Tl	Thallium	81
201	Hg	Mercury	80
197	Au	Gold	79
195	Pt	Platinum	78
192	Ir	Iridium	77
190	Os	Osmium	76
186	Re	Rhenium	75
184	W	Tungsten	74
181	Ta	Tantalum	73
179	Hf	Hafnium	72
139	La	Lanthanum	57
227	Ac	Actinium	89

Key

Ar	Symbol	Name	Z
		relative atomic mass	atomic number

Surname	Centre Number	Candidate Number
First name(s)		0

**GCSE****3430UB0-1**

Z22-3430UB0-1

FRIDAY, 17 JUNE 2022 – AFTERNOON**SCIENCE (Double Award)****Unit 2 – CHEMISTRY 1
HIGHER TIER****1 hour 15 minutes**

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	9	
2.	6	
3.	6	
4.	10	
5.	6	
6.	6	
7.	8	
8.	9	
Total	60	

3430UB01
01**ADDITIONAL MATERIALS**

In addition to this examination paper you will need a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid. You may use a pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question **5** is a quality of extended response (QER) question where your writing skills will be assessed.

The Periodic Table is printed on the back cover of this paper and the formulae for some common ions on the inside of the back cover.



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Answer **all** questions.

1. (a) (i) State why sodium is stored in oil in the laboratory. [1]

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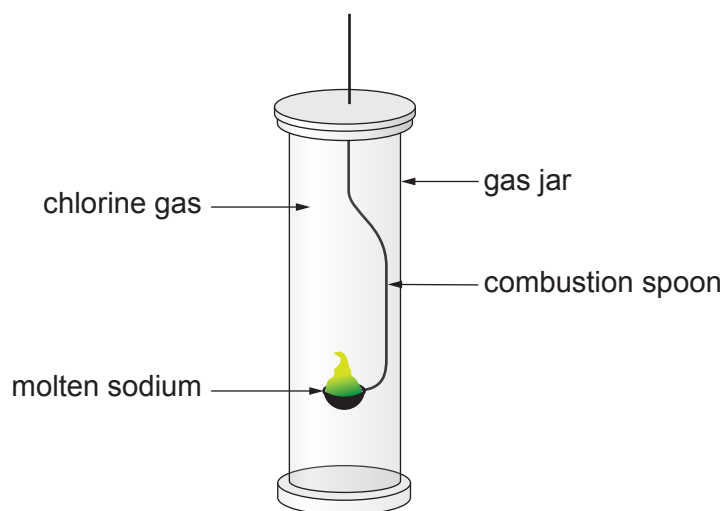
- (ii) Describe the change in appearance when a piece of freshly cut sodium is left for a few minutes. [1]

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- (iii) Give the formula of the compound formed when sodium reacts with oxygen. [1]

.....

- (b) The diagram shows the reaction of sodium with chlorine.



- (i) State why it is necessary to carry out this reaction in a fume cupboard. [1]

.....

- (ii) Complete and balance the equation for the reaction of sodium and chlorine. [2]



(c) The table shows some properties of Group 7 elements.

Element	Melting point (°C)	Boiling point (°C)	Reaction with hot iron
fluorine	−220	−188	explosive
chlorine	−101	−34	very fast
bromine	−7	59	quite fast
iodine	114		slow

(i) Put a tick (✓) in the box next to the most likely boiling point for iodine. [1]

−25 °C ☐ 25 °C ☐ 100 °C ☐ 150 °C ☐

(ii) Astatine lies below iodine in Group 7. State how you would expect astatine to react with hot iron.

Give a reason for your answer.

[2]

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2. (a) In 1915 Alfred Wegener suggested that the Earth's continents were once joined together as one large land mass.



- (i) State **three** pieces of evidence that Alfred Wegener used to support his theory. [3]

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- (ii) Explain why Wegener's theory was not originally accepted by other scientists, but it is today. [2]

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(b) What type of destructive event is likely to happen at a **conservative** plate boundary? [1]

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6

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3. The table shows the electronic structures of atoms of the elements **A–F**.

A–F are not the chemical symbols for the elements.

Element	Electronic structure
A	2
B	2,6
C	2,8,1
D	2,8,7
E	2,8
F	2,8,6

(a) Which of elements **A–F** is found in Group 6 and Period 3 of the Periodic Table?

Explain your choice, referring to electronic structure.

[2]

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.....

(b) Which **two** of elements **A–F** are chemically inert?

Explain your choice, referring to electronic structure.

[2]

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(c) Element **D** has two isotopes. Isotope **1** has 18 neutrons and isotope **2** has 20 neutrons.

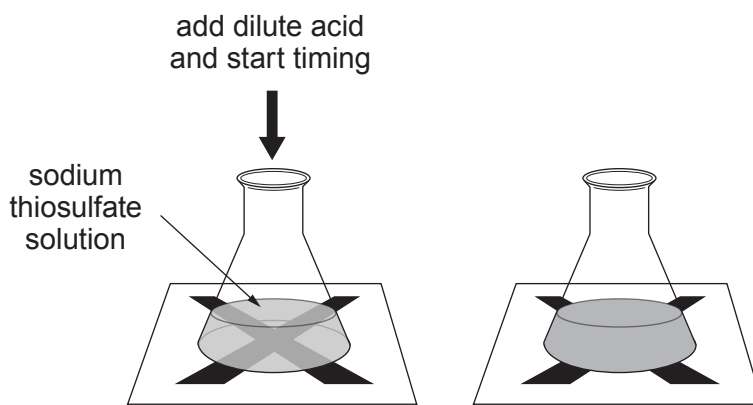
Complete the table by giving the atomic number and mass number of both isotopes. [2]

Isotope	Atomic number	Mass number
1		
2		



4. Sodium thiosulfate solution reacts with dilute hydrochloric acid to form a precipitate. The precipitate causes the solution to go cloudy.

The rate of the reaction can be measured by placing a cross beneath the flask and measuring the time taken for the cross to disappear.



Gareth and Sion studied the effect of sodium thiosulfate concentration by carrying out the reaction with thiosulfate of five different concentrations. They tested each concentration three times.

Their results are shown in the table below.

Concentration of sodium thiosulfate (g/dm ³)	Time 1 (s)	Time 2 (s)	Time 3 (s)	Mean time (s)
0.2	114	113	112	113
0.4	74	70	72	72
0.6	40	38	57	39
0.8	21	23	22	22
1.0	14	16	15	15

- (a) When calculating the mean times, they ignored one of the values recorded.

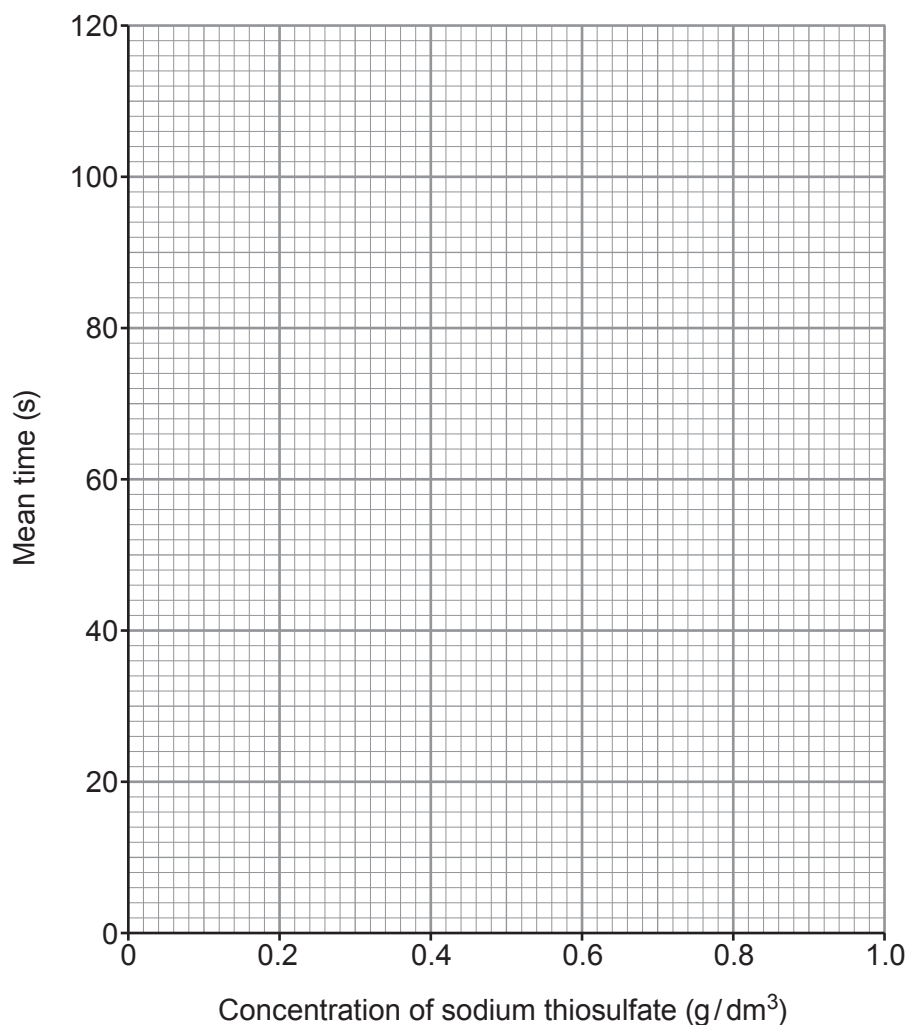
Circle this value in the table.

[1]



- (b) Plot the mean time against sodium thiosulfate concentration on the grid below. Draw a suitable line.

[3]



- (c) Explain the results in terms of particle theory.

[3]

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- (d) A second group of students carried out the same experiment with sodium thiosulfate solution at a higher temperature.

On the grid above, sketch a line to show the results you would expect them to record.

[1]



- (e) The stock solution of sodium thiosulfate used in both experiments was made by dissolving 1.0 g of the solid in 1 dm³ of water.

Calculate the number of moles of sodium thiosulfate (Na₂S₂O₃) in 1.0 g.

Give your answer to **two** significant figures.

[2]

$$A_r(\text{Na}) = 23$$

$$A_r(\text{S}) = 32$$

$$A_r(\text{O}) = 16$$

Number of moles = mol

10



- State the difference between the composition of temporary hard water and permanent hard water. Describe a method to distinguish between them in the laboratory. Explain how this method works.

[6 QER]

3430UB01

6	



6. Levels of oxygen and carbon dioxide in the atmosphere are maintained at approximately constant values.

- (a) Identify the natural processes that help maintain the balance of oxygen and carbon dioxide levels in the atmosphere. Describe briefly how this is achieved. [3]

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- (b) The table shows the amount of carbon dioxide produced by the United States of America and India between 1955 and 2015.

Year	Carbon dioxide emissions (million tonnes)	
	USA	India
1955	700	100
1965	800	250
1975	1150	250
1985	1150	450
1995	1150	600
2005	1400	900
2015	1300	1850

Use the information in the table to compare the increase in carbon dioxide emissions in the United States of America and India between 1955 and 2015. [2]

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- (c) The combustion of the fossil fuel propane (C_3H_8) produces carbon dioxide and water. Balance the equation for this reaction. [1]



7. The table shows some properties of elements in Period 3 of the Periodic Table.

Element	Melting point (°C)	Boiling point (°C)	Density (g/cm ³)	Appearance	Malleability	Conductivity
Na	98	882	1.00	shiny solid	malleable	good
Mg	650	1091	1.75	shiny solid	malleable	good
Al	660	2470	2.70	shiny solid	malleable	good
Si	1410	3265	2.35	shiny solid	brittle	semiconductor
P	44	281	1.80	red solid	brittle	poor
S	113	444	2.05	yellow solid	brittle	poor
Cl	−101	−34	0.003	green gas	n/a	poor

(a) One of the elements is difficult to classify as a metal or non-metal.

Identify this element and give your reasoning.

[2]

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- (b) Tick (✓) **two** boxes which correctly describe the change in density and boiling point for the elements across Period 3. [2]

The density of metals and non-metals increases

☐

The boiling point of metals increases but the boiling point of non-metals shows no trend

☐

The density of metals shows no trend but the density of non-metals decreases

☐

The boiling point of metals and non-metals shows no trend

☐

The density of metals increases but the density of non-metals shows no trend

☐

The boiling point of metals shows no trend but the boiling point of non-metals decreases

☐

The density of metals decreases but the density of non-metals shows no trend

☐

- (c) Argon is the next element in Period 3 after chlorine, Cl.

State why it is not possible to predict a melting point for argon using the information in the table. [1]

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(d) Phosphorus is found in phosphoric acid, H_3PO_4 .

(i) During the production of phosphoric acid, phosphorus is heated to 60°C .

Give the state of phosphorus at 60°C . Explain your choice.

[2]

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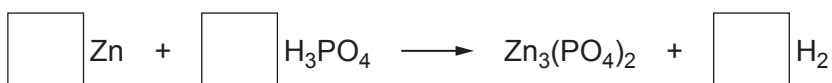
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(ii) Phosphoric acid reacts with zinc to produce zinc phosphate and hydrogen.

Balance the equation for this reaction.

[1]



8



8. The tables show the electronic structures of some Group 1 and Group 2 elements.

Group 1 metal	Electronic structure
sodium	2,8,1
potassium	2,8,8,1

Group 2 metal	Electronic structure
magnesium	2,8,2
calcium	2,8,8,2

(a) (i) Use the information to explain the trend in reactivity down Group 1. [2]

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(ii) Use the information to explain the difference in reactivity of Group 1 and Group 2 elements. [2]

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- (b) Sodium reacts with water to produce sodium hydroxide and hydrogen. The equation for this reaction is shown.



Calculate the mass of sodium needed to produce 11.2g of hydrogen gas. [3]

$$A_r(\text{Na}) = 23 \quad A_r(\text{H}) = 1$$

Mass of sodium = g

- (c) Group 2 metals react in a similar way with water as Group 1 metals.

The word equation for the reaction of calcium and water is shown.



Write the balanced symbol equation for this reaction. [2]

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END OF PAPER



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^-
ammonium	NH_4^+	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^-
calcium	Ca^{2+}	fluoride	F^-
copper(II)	Cu^{2+}	hydroxide	OH^-
hydrogen	H^+	iodide	I^-
iron(II)	Fe^{2+}	nitrate	NO_3^-
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^+	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^+		
silver	Ag^+		
sodium	Na^+		
zinc	Zn^{2+}		





THE PERIODIC TABLE

1 2 3 4 5 6 7 0

Group

1 H Hydrogen 1										4 He Helium 2																									
7 Li Lithium 3		9 Be Beryllium 4		11 B Boron 5		12 C Carbon 6		14 N Nitrogen 7		16 O Oxygen 8		19 F Fluorine 9		20 Ne Neon 10																					
23 Na Sodium 11		24 Mg Magnesium 12		27 Al Aluminium 13		28 Si Silicon 14		31 P Phosphorus 15		32 S Sulfur 16		35.5 Cl Chlorine 17		40 Ar Argon 18																					
39 K Potassium 19		40 Ca Calcium 20		45 Sc Scandium 21		48 Ti Titanium 22		51 V Vanadium 23		52 Cr Chromium 24		55 Mn Manganese 25		56 Fe Iron 26		59 Co Cobalt 27		59 Ni Nickel 28		63.5 Cu Copper 29		65 Zn Zinc 30		70 Ga Gallium 31		73 Ge Germanium 32		75 As Arsenic 33		79 Se Selenium 34		80 Br Bromine 35		84 Kr Krypton 36	
86 Rb Rubidium 37		88 Sr Strontium 38		89 Y Yttrium 39		91 Zr Zirconium 40		93 Nb Niobium 41		96 Mo Molybdenum 42		99 Tc Technetium 43		101 Ru Ruthenium 44		103 Rh Rhodium 45		106 Pd Palladium 46		108 Ag Silver 47		112 Cd Cadmium 48		115 In Indium 49		119 Sn Tin 50		122 Sb Antimony 51		128 Te Tellurium 52		127 I Iodine 53		131 Xe Xenon 54	
133 Cs Caesium 55		137 Ba Barium 56		139 La Lanthanum 57		179 Hf Hafnium 72		181 Ta Tantalum 73		184 W Tungsten 74		186 Re Rhenium 75		190 Os Osmium 76		192 Ir Iridium 77		195 Pt Platinum 78		197 Au Gold 79		201 Hg Mercury 80		204 Tl Thallium 81		207 Pb Lead 82		209 Bi Bismuth 83		210 Po Polonium 84		210 At Astatine 85		222 Rn Radon 86	
223 Fr Francium 87		226 Ra Radium 88		227 Ac Actinium 89																															
														Key																					

Key

relative atomic mass

Ar

Symbol

Name

Z

atomic number

Surname	Centre Number	Candidate Number
First name(s)		0



GCSE
3430UB0-1



FRIDAY, 16 JUNE 2023 – MORNING

SCIENCE (Double Award)
Unit 2 – CHEMISTRY 1
HIGHER TIER
1 hour 15 minutes

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	8	
2.	7	
3.	4	
4.	6	
5.	8	
6.	9	
7.	9	
8.	9	
Total	60	

ADDITIONAL MATERIALS

In addition to this examination paper you will need a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

You may use a pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question **4** is a quality of extended response (QER) question where your writing skills will be assessed.

The Periodic Table is printed on the back cover of this paper and the formulae for some common ions on the inside of the back cover.

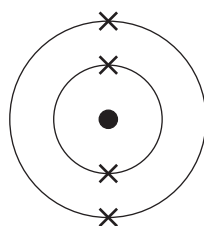


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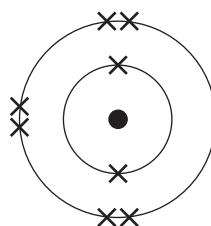
Answer **all** questions.

1. The diagrams below show the electronic structures of five elements, **A**, **B**, **C**, **D** and **E**.

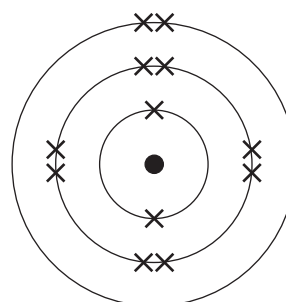
The letters are not the symbols of the elements.



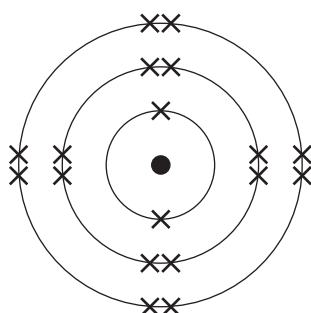
A



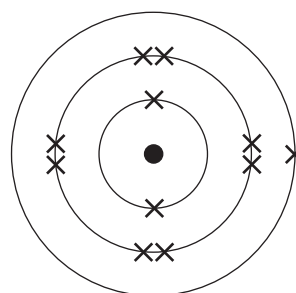
B



C



D



E

- (a) Give the **letters** of the elements found in Period 2 of the Periodic Table. Give a reason for your choice. [2]

Letters and

Reason

.....



- (b) Give the **letter** of the element found in Group 0 of the Periodic Table. Give a reason for your choice. [2]

Letter

Reason

- (c) Explain how the electronic structure of element **E** can be used to determine its atomic number. [2]

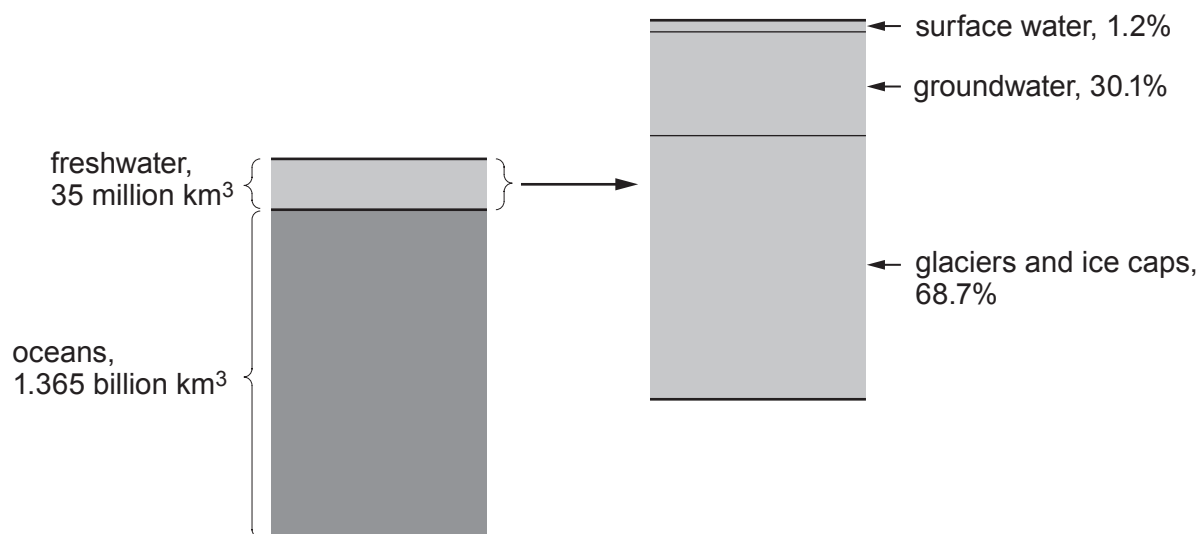
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- (d) One of the diagrams represents the element oxygen. Oxygen reacts with potassium to form potassium oxide.

Give the formula for potassium oxide and balance the equation for this reaction. [2]



2. The diagram shows the distribution of water on Earth.



- (a) Use the information in the diagram to calculate the volume of freshwater that is available as surface water. [2]

Volume = million km³

- (b) As the Earth's population continues to increase, there is going to be a greater need to get freshwater from seawater. This is known as desalination.

Give a method that could be used to obtain freshwater from seawater. [1]

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- (c) Calcium ions, Ca^{2+} , cause hardness in groundwater.

Describe an experiment to **compare** the hardness of two groundwater samples. [3]

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- (d) Washing soda is used to soften hard water. This contains sodium carbonate.

Give the formula of sodium carbonate. [1]

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3430UB01
05

7



3. A food colouring contains a mixture of yellow and green dyes.

The table shows the R_f values of these dyes using different solvents in separate chromatography experiments.

Colour	R_f value when the solvent is		
	Water	Acid	Alcohol
yellow	0.74	0.35	0.00
green	0.76	0.81	0.46

- (a) Why would yellow and green dyes be difficult to separate in a chromatography experiment using **water** as the solvent?

[1]

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- (b) Describe what you would expect to see in the chromatogram obtained using **alcohol** as the solvent.

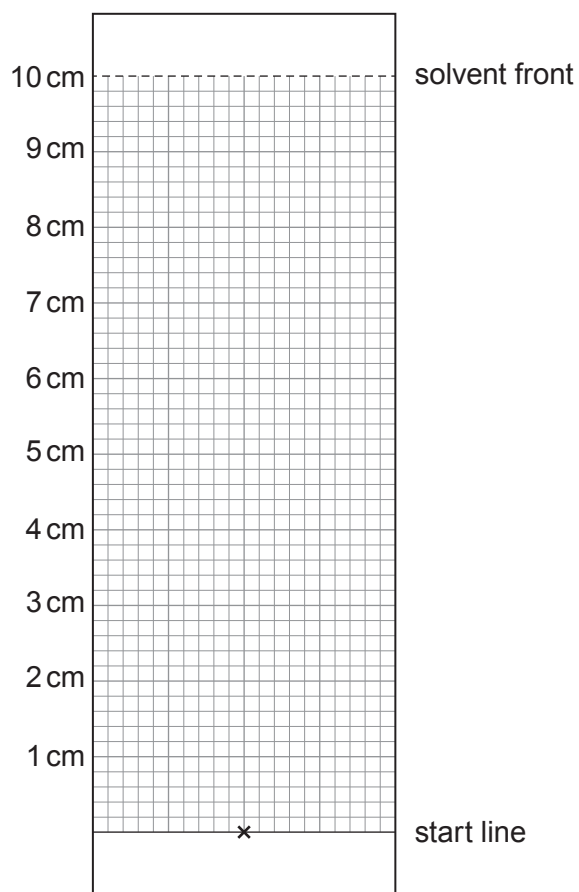
[1]

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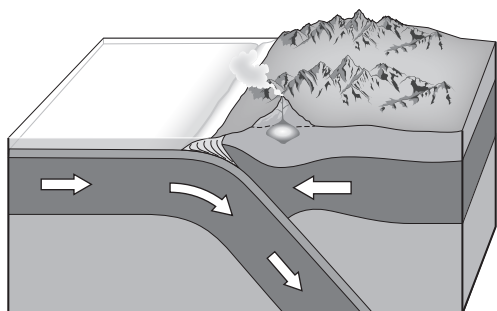


- (c) Complete the chromatogram to show the results you would expect using **acid** as the solvent. [2]

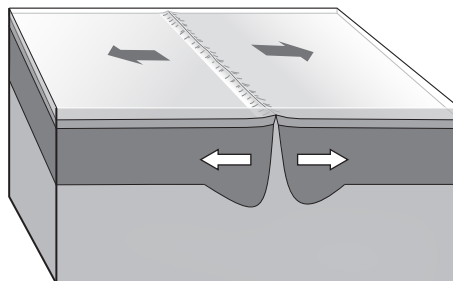


Examiner
only

4. The diagrams show two types of plate boundaries, **A** and **B**. Describe what happens at each boundary. [6 QER]



A

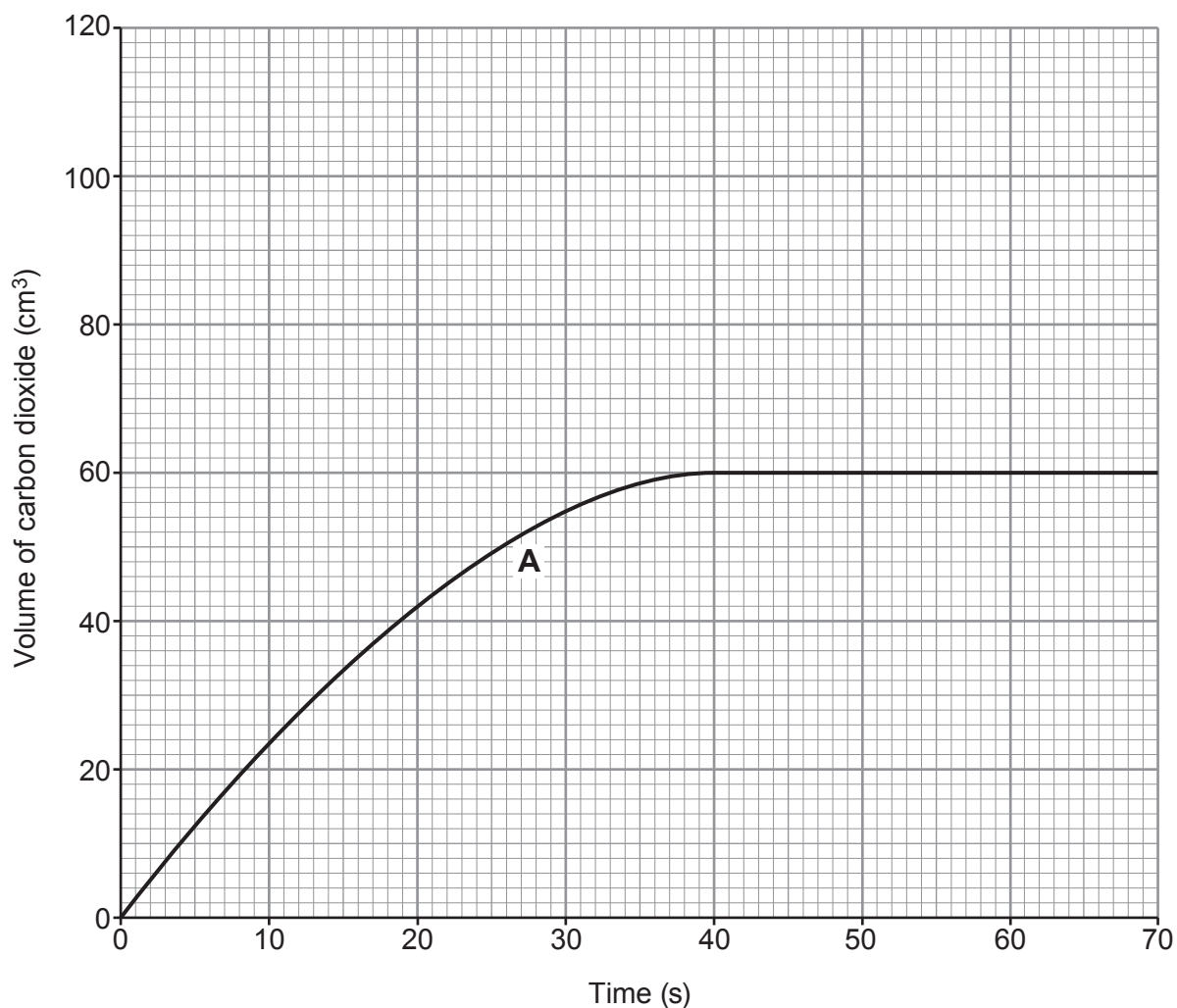


B



5. Marble chips (calcium carbonate) react with dilute hydrochloric acid to produce carbon dioxide gas.

Graph **A** shows the volume of carbon dioxide produced during the reaction between 0.25 g of marble chips and **excess** dilute hydrochloric acid at 30 °C.



- (a) State why no more marble chips remain when the reaction stops. [1]

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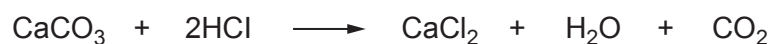
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- (b) The original experiment was repeated at 60 °C.

On the grid draw the curve you would expect to see for the production of carbon dioxide at this temperature. [1]



- (c) The equation for the reaction between calcium carbonate and dilute hydrochloric acid is as follows.



Calculate the mass of calcium chloride produced when 7.8 g of calcium carbonate is reacted with excess acid. [2]

$$M_r(\text{CaCO}_3) = 100$$

$$M_r(\text{CaCl}_2) = 111$$

Mass = g

- (d) When the reaction was repeated with a different mass of calcium carbonate, it was found that 14.3 g of calcium chloride formed. This represents a yield of 53.7%.

Calculate the maximum mass of calcium chloride that could have been formed. [2]

Maximum mass = g

- (e) Catalysts can be added to some chemical reactions in order to increase their rate.

Explain how a catalyst increases the rate of a chemical reaction. [2]

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6. Fluoride ions are added to drinking water in some areas of the UK.

The table below shows the effect of different levels of fluoride in drinking water on the number of decayed, missing and filled teeth (DMFT) and the percentage of people suffering from fluorosis.

Concentration of fluoride (mg/dm ³)	Mean DMFT	Percentage of people affected by fluorosis (%)
0.3	7.0	4
0.6	4.5	6
0.9	3.0	15
1.2	2.5	35
1.5	2.3	40
1.8	2.0	45
2.1	2.1	60
2.4	2.1	68
2.7	2.1	75

- (a) The recommended concentration of fluoride ions to be added to drinking water in the UK is 1.0 mg/dm³.

Use the information in the table to explain why this is the recommended level in terms of DMFT and fluorosis. [4]

DMFT

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.....

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Fluorosis

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.....

.....



- (b) Chlorine is added to all drinking water in the UK, whilst fluoride is only added in some areas.

Explain why some people are opposed to adding fluoride but no-one is opposed to adding chlorine to drinking water. [2]

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- (c) Fluorine reacts with titanium at very high temperatures, forming titanium fluoride.

In an experiment 3.8 g of fluorine reacted with 2.4 g of titanium.

Calculate the simplest formula for titanium fluoride. You **must** show your working. [3]

$$A_r(\text{Ti}) = 48$$

$$A_r(\text{F}) = 19$$

Simplest formula

9



7. A great deal of scientific research into global warming has happened over the last 30 years. More recently, two new contrasting theories linked to global warming have been widely reported. These are outlined below.

Solar activity

A small number of scientists believe that changes in solar activity (changes in the brightness and warmth of the Sun) is causing global warming. Solar activity occurs on the surface of the Sun when nuclear reactions cause flares to be released, high winds to occur and a release of high energy particles from the Sun.

Global dimming

Some scientists believe that global warming is increasing the Earth’s mean temperature, but that global dimming is preventing it from increasing even more. Global dimming is the reduction in the amount of radiation reaching the Earth’s surface from the Sun. It is thought to be caused by an increase of aerosol particles in the Earth’s atmosphere caused by pollution, dust and smog, as well as volcanic eruptions. These particles absorb solar energy and reflect sunlight back into space.

Figure 1 shows the mean atmospheric temperature, carbon dioxide concentration in the atmosphere and solar activity between 1930 and 2010.

Figure 1

Year	Mean atmospheric temperature (°C)	CO ₂ concentration (ppm)	Solar activity (arbitrary units)
1930	13.6	309	3.9
1940	13.6	311	4.0
1950	13.8	316	4.2
1960	13.9	320	4.2
1970	14.0	331	4.0
1980	14.2	342	3.9
1990	14.3	358	3.7
2000	14.4	370	3.7
2010	14.6	383	3.6

ppm = parts per million



Figure 2 shows the mean atmospheric temperature within a 20-mile radius of the Mount Pinatubo volcano in the Philippines between 1990 and 2000. Mount Pinatubo erupted in 1991.

Figure 2

Year	Mean atmospheric temperature (°C)
1990	15.1
1991	14.7
1992	14.8
1993	14.7
1994	14.7
1995	14.6
1996	14.7
1997	14.8
1998	14.7
1999	14.8
2000	14.9

- (a) Tick (✓) the box that **best** describes the change in carbon dioxide concentration in the atmosphere between 1930 and 2010. [1]

Carbon dioxide concentration increased by approximately 10 ppm every 10 years

☐

Carbon dioxide concentration increased more between 1970 and 2010 than it did between 1930 and 1960

☐

Carbon dioxide concentration increased more between 1930 and 1960 than it did between 1970 and 2010

☐

There is no trend to the change in carbon dioxide concentration between 1930 and 2010

☐


- (b) A newspaper report states that global warming is caused by increases in atmospheric carbon dioxide levels **and** by increases in solar activity.

Explain whether the data in **Figure 1** supports this claim.

[2]

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- (c) **Figure 2** shows that the mean atmospheric temperature near Mount Pinatubo decreased after the volcanic eruption of 1991. Suggest why scientists would **not** use these findings to explain how volcanic activity affects **global** temperatures.

[1]

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- (d) State **one** industrial method that is being developed to reduce the concentration of carbon dioxide in the atmosphere.

[1]

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- (e) The Earth's original atmosphere contained about 95% carbon dioxide. Today, the atmosphere contains 0.04% carbon dioxide. Explain why the level of carbon dioxide decreased over geological time.

[3]

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- (f) When carbon dioxide turns limewater milky, calcium hydroxide is produced.

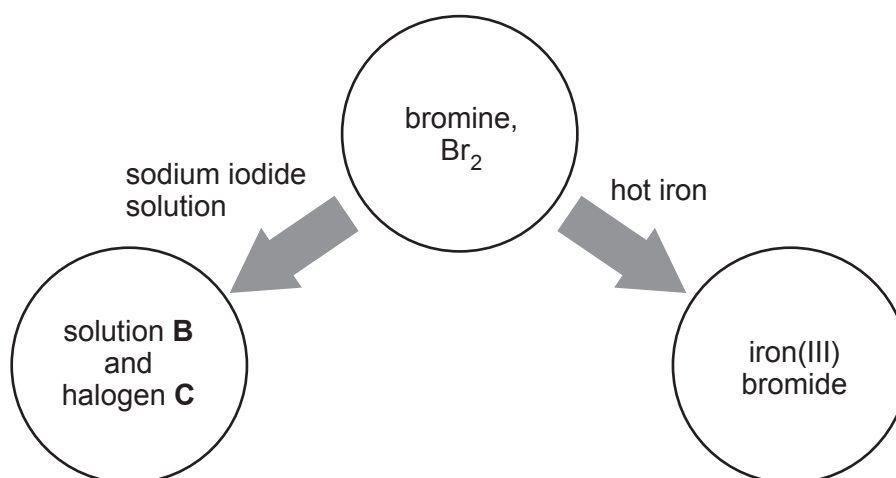
Give the formula of calcium hydroxide.

[1]

.....



8. The diagram shows reactions of bromine, Br_2 .



- (a) (i) Give the names of solution **B** and halogen **C**. [2]

solution **B**

halogen **C**

- (ii) Explain why bromine reacts with sodium iodide solution. [2]

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- (b) Write the symbol equation for the reaction between bromine, Br_2 , and iron to form iron(III) bromide. [2]

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- (c) Silver nitrate solution was added to a solution of sodium iodide. Silver iodide was formed.

- (i) State what you would expect to see. [1]

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- (ii) Give the **ionic** equation for the formation of silver iodide. [2]

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END OF PAPER



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^-
ammonium	NH_4^+	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^-
calcium	Ca^{2+}	fluoride	F^-
copper(II)	Cu^{2+}	hydroxide	OH^-
hydrogen	H^+	iodide	I^-
iron(II)	Fe^{2+}	nitrate	NO_3^-
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^+	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^+		
silver	Ag^+		
sodium	Na^+		
zinc	Zn^{2+}		





THE PERIODIC TABLE

1 2 3 4 5 6 7 0

Group

1 H Hydrogen 1																						
7 Li Lithium 3		9 Be Beryllium 4																				
23 Na Sodium 11		24 Mg Magnesium 12																				
39 K Potassium 19		40 Ca Calcium 20		45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	63.5 Cu Copper 29	65 Zn Zinc 30									
86 Rb Rubidium 37		88 Sr Strontium 38		89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	99 Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	127 I Iodine 53	131 Xe Xenon 54				
133 Cs Caesium 55		137 Ba Barium 56		139 La Lanthanum 57	179 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86			
223 Fr Francium 87		226 Ra Radium 88		227 Ac Actinium 89																		

Key

1
H
Hydrogen
1

Key

A_r

Symbol

Name

Z

relative atomic mass

atomic number